
FLEXIBLE JOB SHOP SCHEDULING PROBLEM WITH TIME WINDOWS

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ABSTRACT

1 Introduction

Community group buying has rapidly gained popularity as an e-commerce model, offering consumers a convenient way to purchase fresh products through online platforms, with the option to pick up items at designated locations the following day. This model streamlines the supply chain by reducing intermediaries, which lowers product prices and improves cost-effectiveness. It has become especially popular among urban consumers in China, addressing the demand for both convenience and affordability. The supply chain in community group buying consists of three main components: warehouses, distributors, and self-pickup points. Warehouses process and sort goods from suppliers according to distributor needs, and ship them to distributors on time. Distributors then perform secondary sorting before delivering goods to self-pickup points for consumer collection. Efficient warehouse operations are critical to meeting tight delivery deadlines and minimizing operational costs. Despite the advantages, the scheduling of warehouse operations presents several challenges. Goods in the warehouse must undergo different processing steps (e.g., shelving, processing, sorting), and resources such as sorting docks and processing lines are limited. Moreover, suppliers deliver goods in multiple batches based on predicted demand, leading to dynamic and unpredictable arrivals. Each distributor also has personalized delivery requirements based on order volume, product type, and the distance to self-pickup points. These complexities make it difficult to optimize warehouse schedules while minimizing production costs and ensuring timely deliveries.

Thus, this research focuses on optimizing warehouse scheduling in community group buying, addressing these challenges by modeling the problem as a flexible job shop scheduling problem (FJSP). The aim is to develop a solution that minimizes cost and maximizes efficiency, considering dynamic deliveries, limited processing resources, and varying delivery requirements.

1.1 Motivation

The rapid emergence of *New Retail*—an innovative commerce model that integrates online platforms with offline fulfillment infrastructures—has significantly transformed traditional supply chains into highly responsive, data-driven ecosystems. In this paradigm, customer demands are collected in real time through online channels and immediately propagated upstream to manufacturing and distribution units. Consequently, production systems are increasingly required to operate under dynamic and uncertain conditions, where job requests do not follow a fixed schedule but instead arrive sporadically and unpredictably in response to volatile consumer behavior.

This shift presents profound challenges to conventional production scheduling methods, which are typically designed for static environments with complete information known in advance. In particular, the emergence of **dynamic job arrivals** necessitates scheduling models that are capable of real-time responsiveness. In community group-buying and on-demand fulfillment platforms—key application scenarios under New Retail—consumer orders are often consolidated

within short time windows and subsequently transmitted to central warehouses or manufacturing centers for rapid execution. These production jobs are then processed and dispatched to geographically distributed distributors, all under tight time constraints and strict delivery commitments.

To meet these requirements, central warehouse operations must simultaneously address two tightly coupled decision-making stages: (i) the scheduling of incoming production jobs on a set of flexible machines, and (ii) the batching and dispatching of completed jobs to external distributors. The complexity of this *Integrated Flexible Job Shop Scheduling and Dispatching Problem with Dynamic Job Arrivals* (IFJSSP-DP-DJA) stems not only from the combinatorial nature of each subproblem, but also from the need to integrate them under real-time, incomplete, and evolving information flows.

Traditional optimization-based scheduling techniques often struggle to provide timely solutions in such dynamic settings due to their computational complexity and lack of adaptability. In contrast, advanced learning-based methods—particularly those grounded in **deep reinforcement learning (DRL)**—offer promising capabilities for reactive decision-making, scalability, and policy generalization in uncertain environments. This motivates the development of novel scheduling frameworks that can continuously learn and adapt under the evolving conditions of New Retail-driven production-distribution systems.

2 Literature Review

3 Problem Formulation

This study addresses the Integrated Flexible Job Shop Scheduling and Dispatching Problem under Dynamic Job Demand (IFJSSP-DP-DJD), which arises in production environments where a set of jobs arrives dynamically over time. These jobs must be processed and subsequently dispatched to multiple distributors, subject to specific delivery requirements.

The concept of dynamic job demand refers to a scenario where job requests are driven by real-time consumer orders, resulting in unpredictable and time-varying job arrivals. Unlike traditional batch-oriented production systems, where job arrivals are generally predetermined, dynamic job demand introduces significant uncertainty as jobs are continuously generated by customer demands, often with varying processing times and delivery constraints. This situation is particularly prevalent in the context of modern retail business models, where online demand is tightly integrated with offline fulfillment systems. Consumer orders are transmitted in near real-time to upstream production systems, creating a dynamic flow of job requests that are unpredictable throughout the scheduling horizon.

The central challenge of this problem lies in the simultaneous optimization of two interdependent decision stages:

Job Scheduling in a Flexible Job Shop: In the processing stage, each job consists of a predefined sequence of operations. These operations can be processed on multiple alternative machines, with machine-dependent and deterministic processing times. However, the exact arrival times of jobs are not known in advance and are subject to dynamic fluctuations. This necessitates a responsive and adaptive scheduling strategy that can make real-time decisions as new jobs arrive. Key decisions include assigning operations to machines, sequencing operations while adhering to precedence constraints, and ensuring machine exclusivity and capacity limitations.

Job Dispatching and Batching: Upon completion of processing, jobs must be grouped into batches and dispatched to designated distributors. Each batch incurs a fixed base loading time and an additional variable time proportional to its size. A batch can only be dispatched once all jobs within it are fully processed and loaded. Furthermore, each job is uniquely assigned to one batch for its corresponding distributor. Distributors have specific delivery requirements, including a required completion percentage of jobs and a due date by which the batch must be dispatched. Tardiness in meeting these delivery requirements incurs a penalty, with the penalty weight reflecting the importance of each requirement.

The objective of this study is to develop an integrated scheduling and dispatching plan that minimizes the total weighted tardiness of all delivery requirements. The dynamic nature of job demand—driven by the variability in end-user demand and the real-time requirements of order fulfillment—introduces temporal uncertainty into the system. This uncertainty necessitates the development of a flexible, event-driven scheduling framework capable of incrementally updating schedules in real time to ensure timely job fulfillment and efficient resource utilization.

3.1 Mathematical Model

This section details the mathematical model for the integrated flexible job shop scheduling and dispatching problem. The model aims to minimize the total weighted tardiness of deliveries by coordinating processing operations and dispatching activities.

The indices and sets used in the model are defined in Table 1.

Table 1: Indices and Sets

Set Notation	Description
J	Set of all jobs to be processed.
M	Set of all available machines.
O_j	Sequence of operations for job $j \in J$. Let $ O_j $ be the number of operations for job j .
$A_{j,o}$	Set of available machines for operation o of job j .
D	Set of all distributors.
I_d	Set of jobs to be dispatched to distributor $d \in D$.
B_d	Set of available dispatch batches for distributor $d \in D$.
R	Set of all delivery requirements.

The parameters used in the model are defined in Table 2.

Table 2: Parameters

Symbol	Stage	Description
$P_{j,o,m}$	Processing	Processing time of operation o of job j on machine m .
T_{base}	Dispatching	Base loading time for a dispatch batch.
T_{item}	Dispatching	Loading time per item in a dispatch batch.
ρ_r	Dispatching	For delivery requirement $r \in R$: required completion percentage (e.g., 0.9 for 90%).
δ_r	Dispatching	For delivery requirement $r \in R$: due date.
w_r	Dispatching	For delivery requirement $r \in R$: tardiness weight.

The decision variables used in the model are defined in Table 3.

Table 3: Decision Variables

Symbol	Type / Domain	Description
<i>Processing Stage Variables</i>		
$x_{j,o,m}$	Binary $\in \{0, 1\}$	1 if operation o of job j is assigned to machine m ; 0 otherwise.
$s_{j,o}$	Continuous ≥ 0	Start time of operation o of job j .
$e_{j,o}$	Continuous ≥ 0	End time of operation o of job j .
$z_{j,o,j',o',m}$	Binary $\in \{0, 1\}$	(Implicit in notebook, for Big-M) 1 if operation (j, o) precedes (j', o') on machine m ; 0 if (j', o') precedes (j, o) on machine m .
<i>Dispatching Stage Variables</i>		
$y_{i,d,b}$	Binary $\in \{0, 1\}$	1 if job i (assuming $i \in J$) is assigned to batch b for distributor d ; 0 otherwise.
$t_{d,b}$	Continuous ≥ 0	Dispatch time of batch b for distributor d .
τ_r	Continuous ≥ 0	Tardiness for delivery requirement r .
$q_{r,b}$	Binary $\in \{0, 1\}$	1 if batch b is used to fulfill delivery requirement r ; 0 otherwise.

The objective is to minimize the total weighted tardiness of all delivery requirements:

$$\min \sum_{r \in R} w_r \tau_r \quad (1)$$

$$e_{j,o} \leq s_{j,o+1} \quad \forall j \in J, o \in O_j \setminus \{|O_j| - 1\} \quad (2)$$

$$\sum_{m \in A_{j,o}} x_{j,o,m} = 1 \quad \forall j \in J, o \in O_j \quad (3)$$

$$e_{j,o} = s_{j,o} + \sum_{m \in A_{j,o}} P_{j,o,m} \cdot x_{j,o,m} \quad \forall j \in J, o \in O_j \quad (4)$$

$$s_{j',o'} \geq e_{j,o} - K \cdot (3 - x_{j,o,m} - x_{j',o',m} - z_{j,o,j',o',m}) \quad \forall (j,o) \neq (j',o'), m \quad (5)$$

$$s_{j,o} \geq e_{j',o'} - K \cdot (2 - x_{j,o,m} - x_{j',o',m} + z_{j,o,j',o',m}) \quad \forall (j,o) \neq (j',o'), m \quad (6)$$

$$\sum_{b \in B_d} y_{i,d,b} = 1 \quad \forall d \in D, i \in I_d \quad (7)$$

$$\sum_{k \in I_d} y_{k,d,b} \leq |I_d| \cdot u_{d,b} \quad \forall d \in D, b \in B_d \quad (8)$$

$$u_{d,b} \leq \sum_{k \in I_d} y_{k,d,b} \quad \forall d \in D, b \in B_d \quad (9)$$

$$t_{d,b} \geq e_{i,|O_i|-1} + T_{\text{base}} \cdot u_{d,b} + T_{\text{item}} \sum_{k \in I_d} y_{k,d,b} - K(1 - y_{i,d,b}) \quad \forall d \in D, b \in B_d, i \in I_d \quad (10)$$

$$\sum_{b \in B_{d_r}} q_{r,b} = 1 \quad \forall r \in R \quad (11)$$

$$q_{r,b} \leq u_{d_r,b} \quad \forall r \in R, b \in B_{d_r} \quad (12)$$

$$\tau_r \geq t_{d_r,b} - \delta_r - K(1 - q_{r,b}) \quad \forall r \in R, b \in B_{d_r} \quad (13)$$

$$s_{j,o}, e_{j,o} \geq 0 \quad \forall j \in J, o \in O_j \quad (14)$$

$$t_{d,b} \geq 0 \quad \forall d \in D, b \in B_d \quad (15)$$

$$\tau_r \geq 0 \quad \forall r \in R \quad (16)$$

The meaning of each formula is as follows: The first formula is the objective function, which minimizes the total weighted tardiness of all delivery requirements. Constraint (2) ensures that the operations of each job are processed in the specified order, so that the start time of the next operation cannot be earlier than the end time of the previous one. Constraint (3) guarantees that each operation is assigned to exactly one available machine. Constraint (4) defines that the end time of each operation equals its start time plus the processing time on the assigned machine. Constraints (5) and (6) use the Big-M method to prevent any two operations from being processed simultaneously on the same machine, thus ensuring machine capacity. Constraint (7) requires that each job for a distributor is assigned to exactly one dispatch batch. Constraints (8) and (9) linearize the indicator variable for whether a batch is used. Constraint (10) ensures that the dispatch time of a batch is not earlier than the completion time of all jobs assigned to it plus the required loading time. Constraint (11) enforces that each delivery requirement is fulfilled by exactly one batch, and constraint (12) ensures that only used batches can be assigned to delivery requirements, thus establishing a one-to-one correspondence between batches and delivery times. Constraint (13) calculates the tardiness for each delivery requirement based on the dispatch time of the corresponding batch and the due date. Finally, constraints (14), (15), and (16) enforce the non-negativity of all time-related variables to ensure the physical feasibility of the model.

3.2 Theoretical Support for Scheduling Policy Design

In this section, we present theoretical results that support the design of reward functions and prioritization strategies in the proposed HDRL scheduling framework. These lemmas are particularly relevant for Chapter 3, where the Integrated Flexible Job Shop Scheduling and Dispatching Problem (IFJSSP-DP) is formulated. The following results provide insights into task sequencing and batch dispatch strategies that help accelerate learning and improve system performance.

Additionally, we consider an extension to the original problem formulation by incorporating uncertainty in job processing times. Specifically, we assume that the actual processing time of each operation is a random variable whose value is not known until the operation begins. This extension reflects practical scenarios such as human-performed tasks,

unpredictable delays in machinery, or variability in product handling, and introduces an additional layer of complexity to the scheduling environment.

In the updated problem setting, let the processing time $P_{j,o,m}$ of operation o for job j on machine m be a random variable, denoted as $\tilde{P}_{j,o,m}$. The realization of $\tilde{P}_{j,o,m}$ is unknown until the operation is initiated. Consequently, the scheduler must make dispatching and sequencing decisions based on historical estimates, expected values $\mathbb{E}[\tilde{P}_{j,o,m}]$, or sampling-based predictions, rather than deterministic values.

This uncertainty affects both stages of the IFJSSP-DP:

- In the processing stage, machine allocation and sequencing must be robust to variability in execution times, potentially relying on adaptive policies that update as real-time feedback becomes available.
- In the dispatching stage, the uncertainty in completion times propagates to uncertainty in dispatch timing, affecting batch formation and due-date compliance. The reward design in HDRL must therefore penalize both high variance and systematic delay.

The following theoretical results still apply in expectation, and guide how the HDRL framework can prioritize low-risk (low-variance) actions when optimizing under uncertainty.

Lemma 1. *Let $\mathcal{J}' = \{J_1, J_2, \dots, J_n\}$ be a set of n jobs ready for processing on a single shared machine, with random processing times $\tilde{p}_j$ having known expectations $\mu_j = \mathbb{E}[\tilde{p}_j]$. Let w_j be the weight associated with job J_j . Then, scheduling the jobs in ascending order of μ_j minimizes the expected total weighted waiting time:*

$$\mathbb{E}[\text{Total Waiting Time}] = \sum_{j=1}^n w_j \cdot \mathbb{E}[W_j]$$

where W_j is the waiting time experienced by job J_j .

Proof. Let π be a job sequence and $W_j^{(\pi)}$ the waiting time for job J_j under π . Since waiting times depend linearly on the sum of preceding job durations, we have:

$$\mathbb{E}[W_j^{(\pi)}] = \sum_{k=1}^{j-1} \mu_{\pi(k)}$$

Thus the total expected weighted waiting time is:

$$\mathbb{E} \left[\sum_{j=1}^n w_j W_j^{(\pi)} \right] = \sum_{j=1}^n w_j \sum_{k=1}^{j-1} \mu_{\pi(k)} = \sum_{1 \leq k < j \leq n} w_j \mu_{\pi(k)}$$

To minimize this sum, we arrange μ_j in ascending order. The proof parallels the deterministic SPT case, extended to expected values. $\square$

Lemma 2. *Let batch b serve delivery requirement r with dispatch time t_b and due date δ_r . Let ρ_r be the required delivery percentage for requirement r , and w_r be the associated tardiness penalty weight. The penalty incurred for tardiness is:*

$$L_b = w_r \cdot \rho_r \cdot \max(0, t_b - \delta_r)$$

Then, L_b is monotonically increasing with ρ_r , assuming all other variables are fixed.

Proof. We analyze the derivative of L_b with respect to ρ_r . Define $\Delta = \max(0, t_b - \delta_r)$, which is non-negative by definition. Then:

$$\frac{\partial L_b}{\partial \rho_r} = w_r \cdot \Delta$$

Since $w_r > 0$ and $\Delta \geq 0$, we conclude that L_b is monotonically increasing with ρ_r . Hence, batches fulfilling denser delivery requirements incur higher potential penalty when delayed. This result justifies reward design strategies that prioritize jobs with high-density delivery constraints. $\square$

Proposition 1. *In the low-level policy learning process of HDRL, incorporating priority-based subgoals (e.g., tagging urgent jobs or vehicles) leads to faster convergence of the policy network under sparse reward conditions.*

Proof. Let π_{std} be a low-level policy trained with standard reward, and π_{shaped} a policy trained with additional subgoal shaping signals for priority tasks. Under sparse reward settings, gradient signals $\nabla J(\theta)$ computed via temporal difference (TD) updates are weak, leading to slow updates in π_{std} .

When priority tasks are tagged and rewarded (shaped), we effectively increase the frequency of reward events, improving the temporal credit assignment. This results in stronger TD error signals, leading to larger and more informative gradient updates:

$$\nabla J(\theta) \propto \mathbb{E}[\delta_t \nabla_{\theta} \log \pi_{\theta}(a_t | s_t)]$$

With reward shaping, δ_t is larger in magnitude and more frequent, thus accelerating convergence. Empirical studies in curriculum learning and reward shaping reinforce this theoretical insight. $\square$

These theoretical results validate the structure of the hierarchical reinforcement learning framework proposed in Chapter 4, and justify key design choices in reward shaping, scheduling prioritization, and dispatch coordination.

4 Solution Methodology

This section outlines the proposed hierarchical reinforcement learning (HRL) framework developed to solve the integrated scheduling and dispatching problem in flexible job shop environments. The solution architecture adopts a two-level decision-making structure, with a meta-level reinforcement learning agent orchestrating the overall control policy and two task-specific agents handling scheduling and dispatching at the operational level. Figure 2 illustrates the hierarchical framework comprising high-level coordination and low-level execution modules.

At the upper level, a meta-controller is implemented as a policy network trained via policy gradient methods. This agent receives a compact nine-dimensional state vector encoding global workshop status—including job urgency, machine load balance, job completion ratio, and scheduling recency—and selects one of three operational modes: (i) scheduling with a Rule-DQN agent, (ii) dispatching via a heuristic batching policy, or (iii) waiting when no feasible action is detected. The meta-agent incorporates an upper confidence bound (UCB) strategy to balance exploration and exploitation, and action masking to enforce operational feasibility.

At the lower level, the framework integrates two distinct action modules. The scheduling module is realized through a hybrid Rule-DQN agent, which combines the domain knowledge of classical dispatching rules with the adaptability of deep Q-learning. Instead of directly outputting scheduling actions, the agent learns to select from a portfolio of eight widely used rules (e.g., SPT, EDD, CR), enabling both interpretability and optimization. Meanwhile, the dispatching module relies on a scoring-based heuristic that prioritizes completed jobs for delivery based on a delay-normalized urgency metric. The batching process groups jobs into delivery sets based on a fixed time window and distributor constraints.

Importantly, unlike conventional bi-level optimization frameworks (e.g., Feng, Che, and Lei 2024; Zhao et al. 2023) or embedded optimization-learning systems (Yan et al. 2024), this framework does not treat the lower-level as a black-box solver invoked by the upper agent. Instead, it models the lower level as an adaptive environment whose response dynamics are learned by the meta-agent over time. This allows the system to flexibly accommodate diverse scheduling and dispatching behaviors, as the upper-level agent implicitly captures the capabilities of lower-level modules through episodic feedback and long-term reward shaping (cf. Su et al. 2023).

Such a design enables dynamic policy switching in response to real-time workshop conditions, supports modularity in integrating domain heuristics and learning agents, and enhances the adaptability and robustness of the overall decision-making system.

4.1 High-Level Agent: Meta-Controller Design

4.1.1 Markov Decision Process Formulation

The decision-making mechanism at the meta-controller level is modeled as a Markov Decision Process (MDP), represented by a tuple $(\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma)$. This formulation enables the high-level agent to interact with the dynamic production-dispatch environment through sequential decision-making. The key components are described as follows:

State ($\mathcal{S}$): Each high-level state $s \in \mathcal{S}$ is represented as a 9-dimensional feature vector that captures the global operational status of the system. These features include the proportion of newly arrived jobs, normalized load imbalance across machines, urgency of pending jobs, time elapsed since the last scheduling decision, current system pressure, job completion and dispatch ratios, average machine utilization, and average waiting time. Formally, the state is defined as:

$$s = [q_{\text{new}}, \sigma_m, \rho_{\text{urgent}}, \tau_{\text{sched}}, \psi, r_{\text{complete}}, r_{\text{dispatch}}, \mu_{\text{util}}, w_{\text{avg}}]^{\top}$$

with each component computed as follows:

$$\begin{aligned}
q_{\text{new}} &= \frac{N_{\text{new}}}{N_{\text{available}}} && \text{(proportion of newly arrived jobs)} \\
\sigma_m &= \frac{\text{StdDev}(\text{remaining_time}_m)}{\text{max_processing_time}} && \text{(normalized load imbalance)} \\
\rho_{\text{urgent}} &= \frac{N_{\text{urgent}}}{N_{\text{available}}} && \text{(ratio of urgent jobs based on due time)} \\
\tau_{\text{sched}} &= \frac{t - t_{\text{last_schedule}}}{\text{max_processing_time}} && \text{(time since last scheduling)} \\
\psi &= \frac{\sum_j p_j}{\sum_m r_m} && \text{(system pressure: total job load / available machine capacity)} \\
r_{\text{complete}} &= \frac{N_{\text{completed}}}{N_{\text{total}}} && \text{(job completion ratio)} \\
r_{\text{dispatch}} &= \frac{N_{\text{dispatched}}}{N_{\text{total}}} && \text{(dispatch completion ratio)} \\
\mu_{\text{util}} &= \frac{1}{M} \sum_{m=1}^M \text{util}_m && \text{(average machine utilization)} \\
w_{\text{avg}} &= \frac{1}{N_{\text{available}}} \sum_{j \in \text{available}} w_j && \text{(average job waiting time)}
\end{aligned}$$

where t is the current time, p_j is the processing time of job j , r_m is the remaining time of machine m , and w_j is the waiting time of job j .

Action ($\mathcal{A}$): The action space consists of three discrete options: $\mathcal{A} = \{\text{schedule}, \text{dispatch}, \text{wait}\}$. At each decision step, the meta-controller selects one of these actions to delegate control to a specialized lower-level module. Each action corresponds to a policy switch that determines how the system proceeds in the current state.

State Transition ($\mathcal{P}$): The transition probability function $\mathcal{P}(s_{t+1} \mid s_t, a_t)$ models the stochastic evolution of the system. Due to dynamic job arrivals, uncertain processing durations, and heterogeneous lower-level policy behaviors, the system transitions from state s_t to s_{t+1} in a probabilistic manner once action a_t is applied. The transition model is not known a priori and must be learned through experience.

Reward Function ($\mathcal{R}$): The high-level reward r_t is designed to provide informative and structured feedback over both scheduling and dispatching outcomes. It combines multiple components:

- **Action-Based Feedback:** A base reward is assigned to each meta-action. Scheduling and dispatching actions are encouraged with positive rewards (e.g., +2.0), while waiting actions are penalized (-1.0) to discourage idling.

Scheduling Quality: When a scheduling action is executed, the reward is defined as:

$$r_t^{\text{schedule}} = \alpha_1 \cdot u_t + \alpha_2 \cdot p_t - \alpha_3 \cdot \sigma_t \quad (17)$$

where u_t is the average machine utilization, p_t is the job progress rate, and σ_t denotes the standard deviation of machine workloads. These metrics are computed as follows:

$$u_t = \frac{1}{M} \sum_{m=1}^M \text{util}_m \quad (18)$$

$$p_t = \frac{N_{\text{completed}}}{N_{\text{total}}} \quad (19)$$

$$\sigma_t = \sqrt{\frac{1}{M} \sum_{m=1}^M (l_m - \bar{l})^2} \quad (20)$$

Here, M is the total number of machines, util_m represents the current utilization of machine m , $N_{\text{completed}}$ and N_{total} denote the number of completed and total operations respectively, l_m is the processing load on machine m , and $\bar{l}$ is the mean load across all machines.

This formulation encourages the agent to prioritize scheduling decisions that increase resource utilization and job throughput while promoting balanced machine workloads.

- **Dispatching Quality:** The dispatching reward is composed of multiple layers of feedback that reflect both job-level and batch-level delivery performance. A base reward is assigned when a dispatch action is selected. Additional shaping rewards are granted when jobs are actively being dispatched or successfully delivered, encouraging continuous dispatch engagement. Specifically, let $\mathbb{I}_{\text{dispatching}}$ and $\mathbb{I}_{\text{dispatched}}$ denote binary indicators for job-level dispatch status, then:

$$r_t^{\text{dispatch}} = \beta_1 \cdot \mathbb{I}_{\text{dispatch_action}} + \beta_2 \cdot \mathbb{I}_{\text{dispatching}} + \beta_3 \cdot \mathbb{I}_{\text{dispatched}} \quad (21)$$

To penalize delayed deliveries, we introduce a tardiness-based deduction. For each completed job j , if its dispatch time d_j exceeds its due date δ_j , a penalty is computed as:

$$r_j^{\text{tardiness}} = -\max(0, d_j - \delta_j) \quad (22)$$

Moreover, distributor-level delivery constraints are modeled through time-windowed delivery ratios. For each delivery requirement r with due time δ_r , delivery ratio ρ_r , total demand A_r , and penalty weight w_r , the following penalty is imposed if delivery falls short:

$$L_r = -w_r \cdot \max(0, \rho_r \cdot A_r - A_r^{\text{delivered}}) \quad (23)$$

These penalties are accumulated over all distributors and subtracted from the total reward. This multi-tiered dispatching reward structure promotes timely job dispatching, batch completeness, and compliance with distributor-specific delivery targets.

- **Shaping Terms:** Positive rewards are given for operation completions, job completions, and successful batch dispatches to reduce reward sparsity and accelerate policy convergence.

To maintain stability, all reward terms are scaled and clipped to prevent divergence, and reward shaping is employed to enable more effective temporal credit assignment during training.

Discount Factor (γ): A discount factor $\gamma \in [0, 1]$ is used to value future rewards in the long-term decision horizon. The objective of the meta-controller is to learn a policy $\pi_\theta(a | s)$ that maximizes the expected cumulative discounted return:

$$J(\theta) = \mathbb{E}_{\pi_\theta} \left[\sum_{t=0}^T \gamma^t r_t \right]$$

This MDP formulation allows the meta-controller to adaptively coordinate between scheduling, dispatching, and passive waiting, enabling responsive and globally aware decision-making in the hierarchical reinforcement learning system.

4.1.2 Network Architecture

The meta-controller is implemented as a fully connected neural policy network with three hidden layers, designed to map a 9-dimensional global feature vector to a probability distribution over three discrete high-level actions: scheduling, dispatching, and waiting. Formally, let $s_t \in \mathbb{R}^9$ denote the input state vector at time t . The forward computation of the network is defined as:

$$h_1 = \text{GeLU}(\text{LayerNorm}(W_1 s_t + b_1)) \quad (24)$$

$$h'_1 = \text{Dropout}(h_1, p = 0.2) \quad (25)$$

$$h_2 = \text{GeLU}(W_2 h'_1 + b_2) \quad (26)$$

$$\text{logits} = W_3 h_2 + b_3 \quad (27)$$

where: - $W_1 \in \mathbb{R}^{32 \times 9}$, $W_2 \in \mathbb{R}^{32 \times 32}$, $W_3 \in \mathbb{R}^{3 \times 32}$ are weight matrices of each layer; - GeLU is the Gaussian Error Linear Unit activation; - LayerNorm is applied after the first layer to stabilize training; - Dropout with probability

$p = 0.2$ is used for regularization; - The final output logits correspond to the action preferences before masking and exploration adjustment.

To enforce feasibility constraints, an action mask $m_t \in \{0, 1\}^3$ is applied to the logits to invalidate infeasible actions:

$$\text{masked_logits}_t = \text{logits}_t + \log m_t, \quad \text{with } \log 0 = -\infty \quad (28)$$

UCB-Based Exploration: To enhance exploration, a UCB-style bonus is added to the logits based on action visit counts $N_t(a)$:

$$\text{UCB}_t(a) = \text{logits}_t(a) + \lambda \cdot \sqrt{\frac{\log T_t}{N_t(a) + \epsilon}}, \quad a \in \{0, 1, 2\} \quad (29)$$

where T_t is the total number of meta decisions so far, λ is the exploration coefficient, and ϵ is a small constant for numerical stability.

The final policy distribution is computed via a softmax over masked and UCB-adjusted logits:

$$\pi_\theta(a_t|s_t) = \frac{\exp(\text{UCB}_t(a_t))}{\sum_{a \in \mathcal{A}} \exp(\text{UCB}_t(a) \cdot m_t(a))} \quad (30)$$

Optimization: The meta-controller is trained using policy gradient with entropy regularization. The loss function is:

$$\mathcal{L}_{\text{meta}}(\theta) = -\mathbb{E}_{\pi_\theta} [\log \pi_\theta(a_t|s_t) \cdot R_t] - \beta \cdot \mathbb{H}[\pi_\theta(\cdot|s_t)] \quad (31)$$

where R_t is the discounted return, β is the entropy regularization coefficient, and $\mathbb{H}$ denotes the Shannon entropy to encourage exploration. Gradients are computed and optimized using Adam with learning rate $\eta = 10^{-3}$ and gradient clipping to $\|\nabla \mathcal{L}\| \leq 1.0$.

Algorithm 1 Meta-Controller Decision and Update Process

- 1: Initialize meta-policy network π_θ , action counts $N(a)$, total steps $T \leftarrow 0$
 - 2: **for** each episode $e = 1, \dots, E$ **do**
 - 3: Reset environment and observe initial global state s_0
 - 4: **for** each decision step $t = 0, \dots, T_{\max}$ **do**
 - 5: Extract feature vector $x_t \in \mathbb{R}^9$ from s_t
 - 6: Compute logits via feedforward network and apply action mask
 - 7: Adjust logits using UCB bonus based on $N(a)$ and T
 - 8: Sample action $a_t \sim \pi_\theta(a_t|x_t)$; update $N(a_t), T$
 - 9: Execute action a_t , observe reward r_t and next state s_{t+1}
 - 10: Store transition $(s_t, a_t, r_t, \log \pi_\theta(a_t), s_{t+1})$ in buffer
 - 11: **if** update condition met **then**
 - 12: Sample batch; compute loss with entropy regularization
 - 13: Update network parameters via policy gradient
 - 14: **if** termination condition met **then break**
-

4.2 Low-Level Actor-Critic Policies

To enhance the learning efficiency and performance robustness of the proposed HDRL framework, we integrate several prioritization heuristics and reward shaping techniques into the low-level policy modules. These heuristics are not arbitrarily chosen but are supported by theoretical insights derived from stochastic scheduling literature and performance-bound analysis.

First, when job processing times are modeled as random variables, we follow the principle that sequencing jobs in ascending order of their expected processing times minimizes the total expected weighted waiting time. This result aligns with the classical SPT rule in deterministic scheduling and provides a theoretical justification for its inclusion in the rule-based action space of the Rule-DQN scheduling agent.

Second, in the dispatching stage, we account for the urgency and density of delivery requirements. Specifically, we show that the penalty incurred for late deliveries increases monotonically with delivery density. Consequently, the scoring function used for dispatch prioritization assigns higher urgency to jobs associated with dense and high-penalty requirements. This design improves the system’s responsiveness to critical deliveries under uncertainty.

Third, under sparse reward conditions, we employ reward shaping strategies to accelerate the convergence of low-level policies. By tagging urgent jobs or vehicles with subgoals, we provide intermediate rewards that enhance gradient signals during temporal difference updates. This strategy has proven effective in accelerating policy learning and stabilizing convergence in HRL settings.

These theoretical results collectively inform the construction of the reward functions, rule sets, and heuristic batching policies adopted in our framework. The proofs and derivations are included for completeness and to reinforce the rationality of our system’s structural design.

The disjunctive graph representation for FJSP is formally defined as $G = (O, C, D)$, where the node set $O = \{O_{ij} \mid \forall i, j\} \cup \{\text{Start}, \text{End}\}$ contains all operations O_{ij} (with i indexing jobs and j indexing operations) plus two dummy nodes $\{\text{Start}, \text{End}\}$ with zero processing time representing production boundaries. The conjunctive arc set C consists of directed arcs forming n acyclic paths from Start to End that encode the predefined processing sequences for each job J_i , while the disjunctive arc set $D = \bigcup_k D_k$ comprises undirected edges where each D_k forms a complete subgraph connecting all operations processable on machine M_k . Notably, in contrast to JSP, any FJSP operation node may belong to multiple machine cliques $\{D_k\}$ due to machine flexibility. The FJSP solution process corresponds to selecting exactly one disjunctive arc per operation and fixing its directionality to establish processing orders on all machines.

We adopt Dueling DQN for the Meta-Controller and actor-critic GATs for the sub-policies. All agents use prioritized experience replay (PER) for sample efficiency. For training stability, target networks are updated using soft updates:

$$\theta' \leftarrow \tau\theta + (1 - \tau)\theta'$$

where τ is the soft update factor. The training loop proceeds episodically, where each episode simulates a full production-distribution horizon with dynamic job arrivals. At each time step, the Meta-Controller observes global state and issues priority and dispatch actions. The resulting sub-goals guide the warehousing and distribution agents, which schedule jobs and deliveries accordingly. All transitions and rewards are stored and sampled for learning using mini-batch TD loss.

The complete training process is summarized in Algorithm 4.2.

5 Computational Experiments

We evaluate the performance of the proposed HDRL framework through extensive computational experiments on benchmark instances of the IFJSSP-DP. The experiments are designed to assess the effectiveness of the hierarchical approach in comparison to traditional methods and to analyze the impact of various design choices.

5.1 Experimental Setup

The experiments are conducted using a simulated production environment that mimics real-world job shop scenarios. We consider a range of problem instances with varying characteristics, including the number of jobs, machines, and resource constraints. The performance metrics used for evaluation include makespan, total tardiness, and resource utilization.

5.2 Results and Discussion

The results of the experiments demonstrate that the HDRL framework outperforms baseline methods in terms of solution quality and adaptability to changing conditions. The hierarchical structure allows for more efficient exploration of the solution space and better handling of complex dependencies between scheduling and dispatching tasks.

6 Conclusion

This paper presents a novel hierarchical reinforcement learning framework for solving the Integrated Flexible Job Shop Scheduling and Dispatching Problem (IFJSSP-DP). By decomposing the problem into high-level coordination and low-level scheduling-dispatching tasks, we leverage the strengths of both rule-based heuristics and deep learning to achieve efficient and adaptive decision-making in complex production environments. The proposed framework

Algorithm 2 HDRL Training Procedure for Warehousing and Distribution Coordination

```

1: Initialize high-level meta-policy Q-network  $\theta_{high}$ , target network  $\theta'_{high}$ , and replay buffer  $B_{high}$ 
2: Initialize low-level sub-policy Q-networks  $\theta_{low}^w, \theta_{low}^d$  (for warehousing and distribution), target networks  $\theta_{low}^{w'}, \theta_{low}^{d'}$ ,
   and replay buffers  $B_{low}^w, B_{low}^d$ 
3: for each episode  $e = 1, \dots, E$  do
4:   Reset simulation environment and obtain initial state  $S_0^{high}, S_0^{low}$ 
5:   for each time step  $t = 0, \dots, T_{max}$  do
6:     High-Level Decision:
7:     Select high-level action  $a_t^{high}$  using  $\epsilon$ -greedy policy on  $Q(S_t^{high}, a; \theta_{high})$ 
8:     Set sub-goals  $g_t^w, g_t^d$  for warehousing and distribution based on  $a_t^{high}$ 
9:     Execute  $a_t^{high}$ , observe intermediate reward  $R_t^{high}$  and next state  $S_{t+1}^{high}$ 
10:    Store transition  $(S_t^{high}, a_t^{high}, R_t^{high}, S_{t+1}^{high})$  in  $B_{high}$ 
11:    Low-Level Decision (Warehousing):
12:    Select action  $a_t^{low,w}$  using  $\epsilon$ -greedy policy on  $Q(S_t^{low,w}, a, g_t^w; \theta_{low}^w)$ 
13:    Execute  $a_t^{low,w}$ , observe reward  $R_t^{low,w}$  and next state  $S_{t+1}^{low,w}$ 
14:    Store transition  $(S_t^{low,w}, a_t^{low,w}, R_t^{low,w}, S_{t+1}^{low,w}, g_t^w)$  in  $B_{low}^w$ 
15:    Low-Level Decision (Distribution):
16:    Select action  $a_t^{low,d}$  using  $\epsilon$ -greedy policy on  $Q(S_t^{low,d}, a, g_t^d; \theta_{low}^d)$ 
17:    Execute  $a_t^{low,d}$ , observe reward  $R_t^{low,d}$  and next state  $S_{t+1}^{low,d}$ 
18:    Store transition  $(S_t^{low,d}, a_t^{low,d}, R_t^{low,d}, S_{t+1}^{low,d}, g_t^d)$  in  $B_{low}^d$ 
19:    Update Networks:
20:    Sample minibatches from  $B_{high}, B_{low}^w, B_{low}^d$  using PER
21:    Perform gradient descent on  $\theta_{high}, \theta_{low}^w, \theta_{low}^d$  to minimize TD loss
22:    Periodically soft-update target networks:  $\theta' \leftarrow \tau\theta + (1 - \tau)\theta'$ 
23:    Update states:  $S_t^{high} \leftarrow S_{t+1}^{high}, S_t^{low} \leftarrow S_{t+1}^{low}$ 
24:    if episode terminates (e.g., all tasks completed or time limit reached) then
25:      Break

```

incorporates theoretical insights from stochastic scheduling literature to inform reward design and action prioritization, enhancing the learning process and system performance. The results demonstrate the effectiveness of the HDRL approach in managing the intricate interactions between scheduling and dispatching, leading to improved overall system performance and adaptability in dynamic environments.